

Lecture 03: Python Ecosystem, Data, and LLMs

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Micron Unveils Micron Research Labs, a U.S.-Based Long-Horizon Innovation Hub to Shape the Future of Memory and AI

AUGUST 20, 2026

Backed by a \$10 billion planned investment over the next decade, Micron Research Labs will unite academia, government, startups and industry to advance the memory and compute breakthroughs that will define the AI era

- **Machine learning** begins with data—it's the foundation upon which models are built. As it evolves, ML increasingly incorporates heuristics, physical laws, complex data structures, and inferential biases, gradually moving from raw data-driven approaches to more sophisticated, context-aware systems.
- **Physics** starts with fundamental principles and laws—these are the bedrock of our understanding of the natural world. From these principles, physics builds models and theories, which are then validated or refined through experimentation and observation, often leading to new discoveries and deeper insights.
- **Mathematics** begins with axioms and definitions—from these basic building blocks, mathematics constructs a logical framework of theorems and proofs. It provides the rigorous language and tools necessary for describing, analyzing, and solving problems across all scientific disciplines, including physics and machine learning

o. Getting big data: making imaging tools a part of data infrastructure

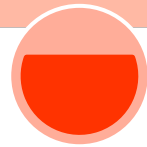
Physics: Why something happens



1. Big data:

How does it happen?

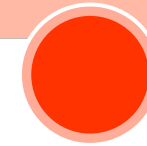
- Unsupervised learning, clustering, and visualization
- **Biggest hurdle:** Language/elementary tools



2. Deep data:

How can we understand?

- Physics informed data analytics/supervised methods
- **Biggest hurdles:** Mathematical framework, scalability of computational tools



3. Smart data: How can we do better?

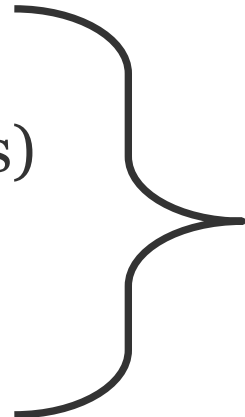
- Feedback and expert/AI systems
- **Biggest hurdles:** With LLMs, it is possible

How it feels most of the time:



Build the case for machine learning

- Explore the business/scientific problem
- Build workflow (operations, costs, latencies)
- Identify bottlenecks
- Chart the solution
- Prototype the solution
- Test and iterate
- Deploy
- Support
- Upgrade
- Sunset



**Homework
assignment 2
(part 1)**

Identify the type of problem

Supervised (inductive) learning

- Given: training data + desired outputs (labels)

• Unsupervised learning

- Given: training data (without desired outputs)

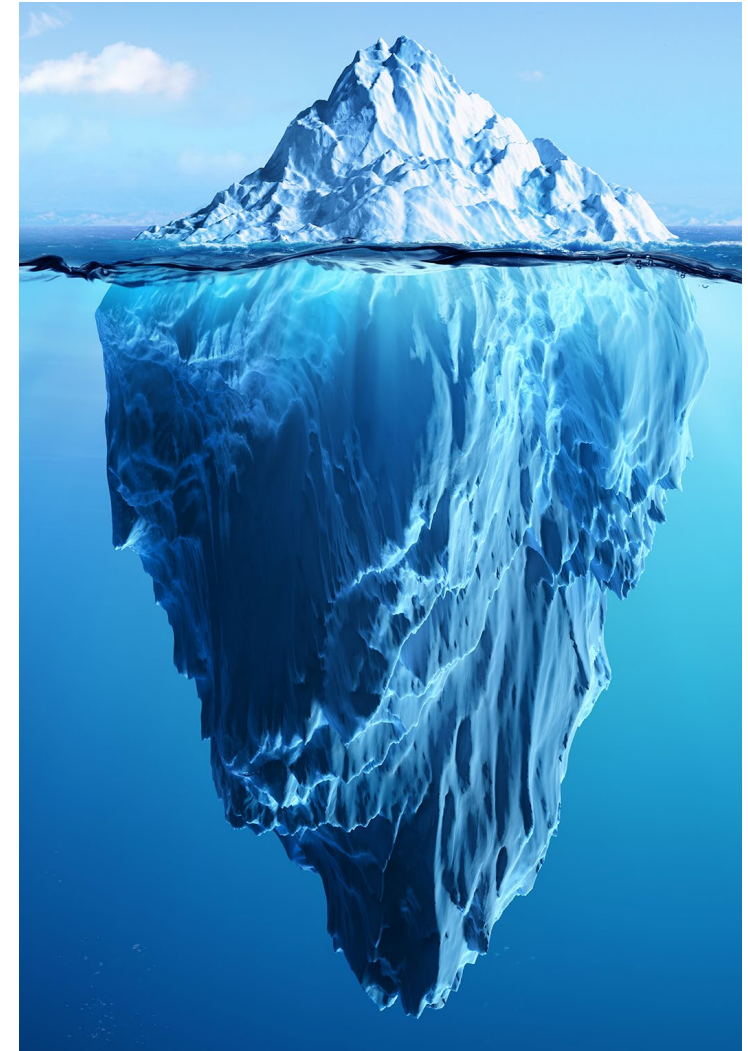
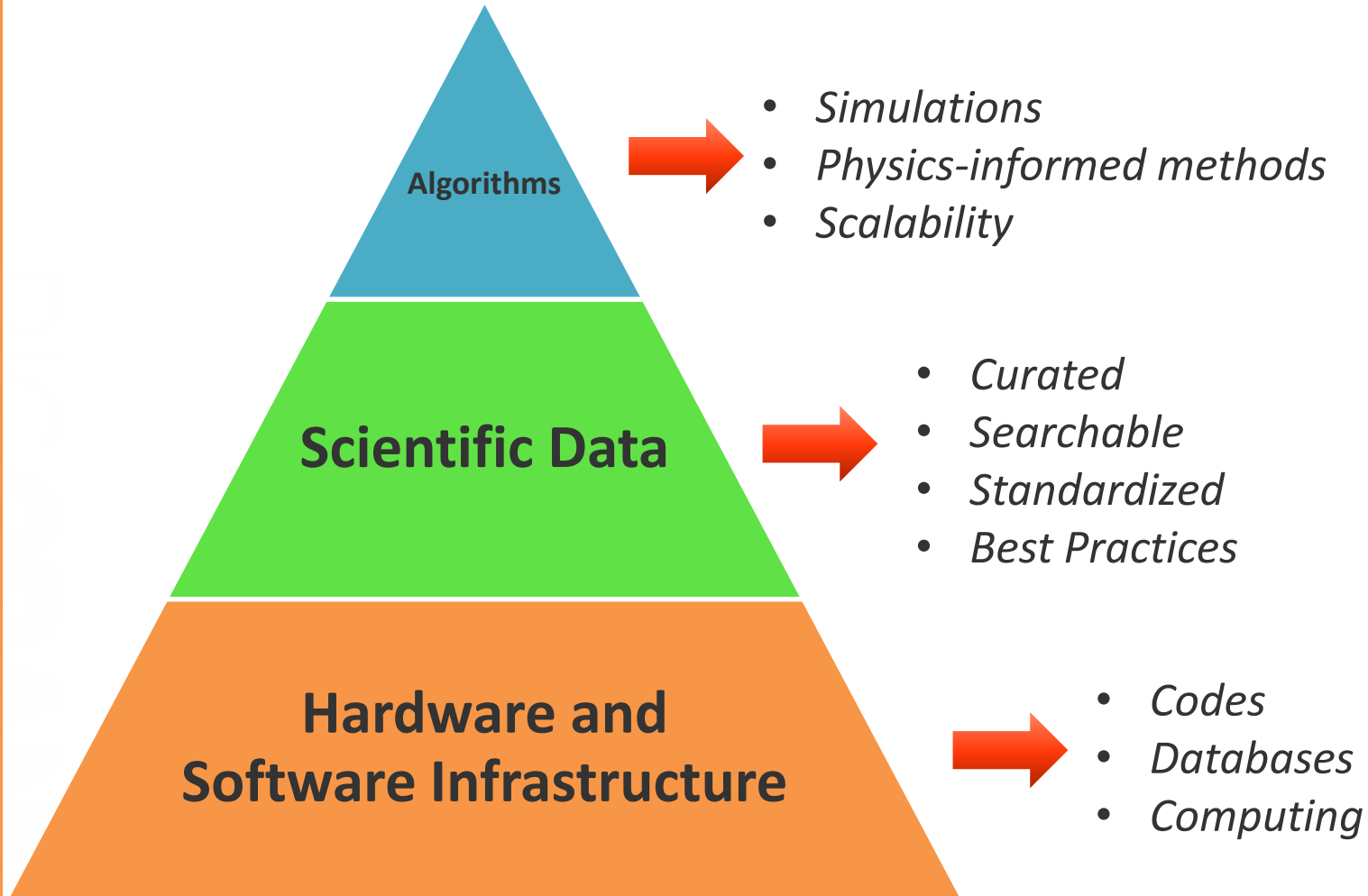
• Semi-supervised learning

- Given: training data + a few desired outputs

• Reinforcement/active learning

- Rewards from sequence of actions

The pyramid of machine learning



Why bother with infrastructure?

- Almost all of machine learning relies extensively on having access to good quality data
- In most laboratories, this data is acquired via multiple instruments in different formats, and not findable or accessible, and often lacks necessary metadata for ML labeling
- As such, in many cases, ML in science is impossible especially in the experimental domains, without the necessary investments in data standardization and storage
- Similarly, reproducibility of workflows relies on strongly tested codebases, not one-off scripts.

Soon to be a requirement (2023)!



- Data management plans in proposals
- Repositories of data and codes with publications
- Good to be ready!

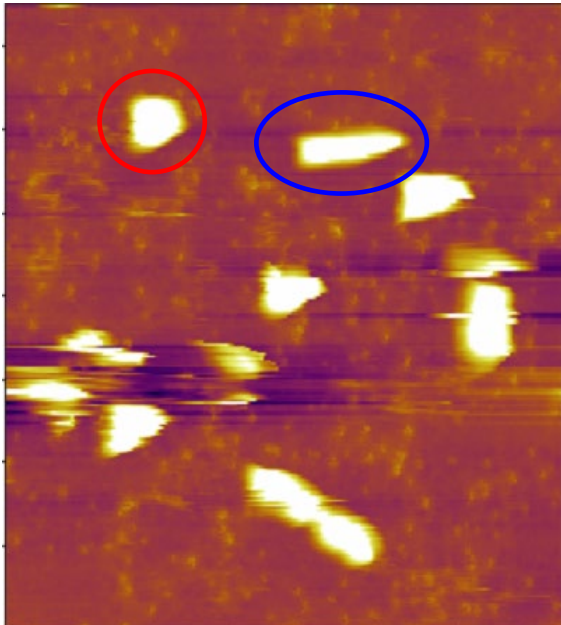
<https://www.science.org/content/article/white-house-requires-immediate-public-access-all-u-s--funded-research-papers-2025>



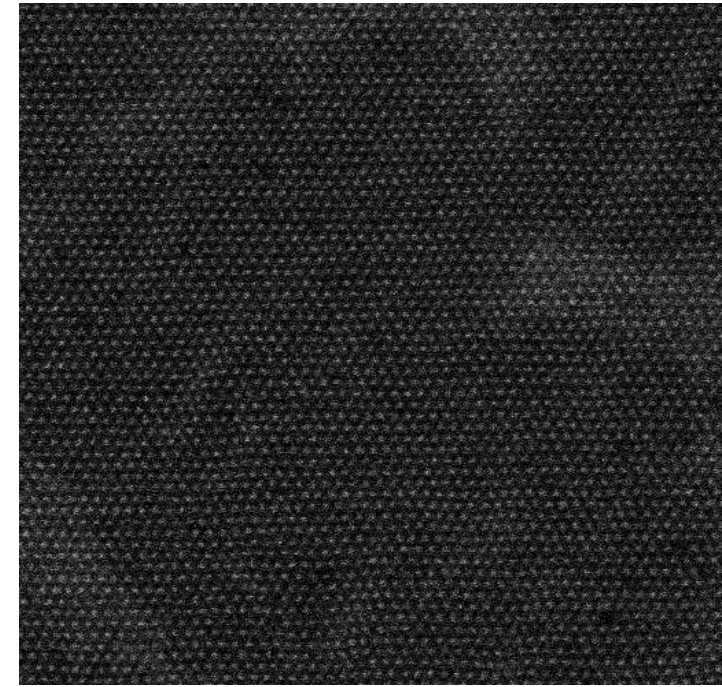
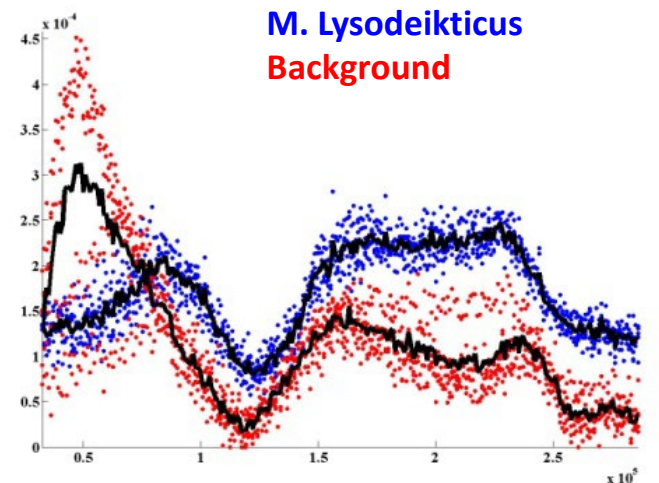
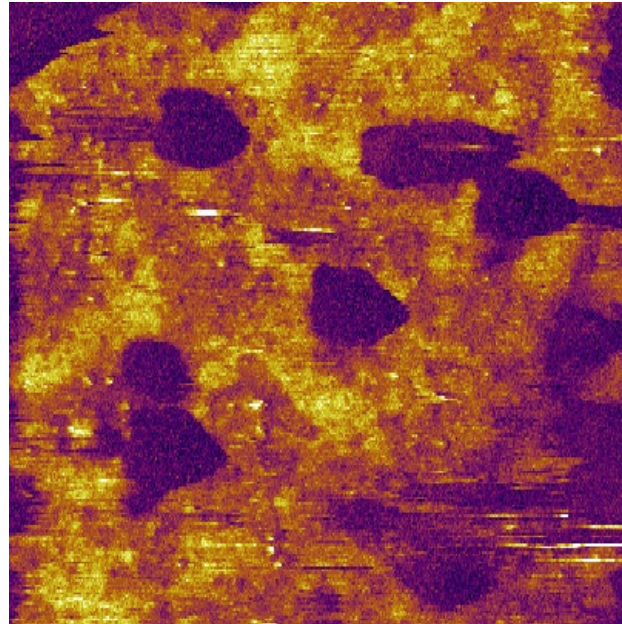
Get data – from scientific tools

- Spectra
- (Multimodal) Images
- Hyperspectral images
- Videos
- Time traces
- ...

Topography



PFM Amplitude



As scientists, we rarely have to deal with the classical ELT (Extract-Load Transform, aka Data Wrangling) problems. But....

Standardization of Microscope Data



Micro Raman Microscope



Atomic Force
Microscope (AFM)



AFM with Infrared
spectroscopy (AFM-IR)



Scanning
Tunneling
Microscope (STM)

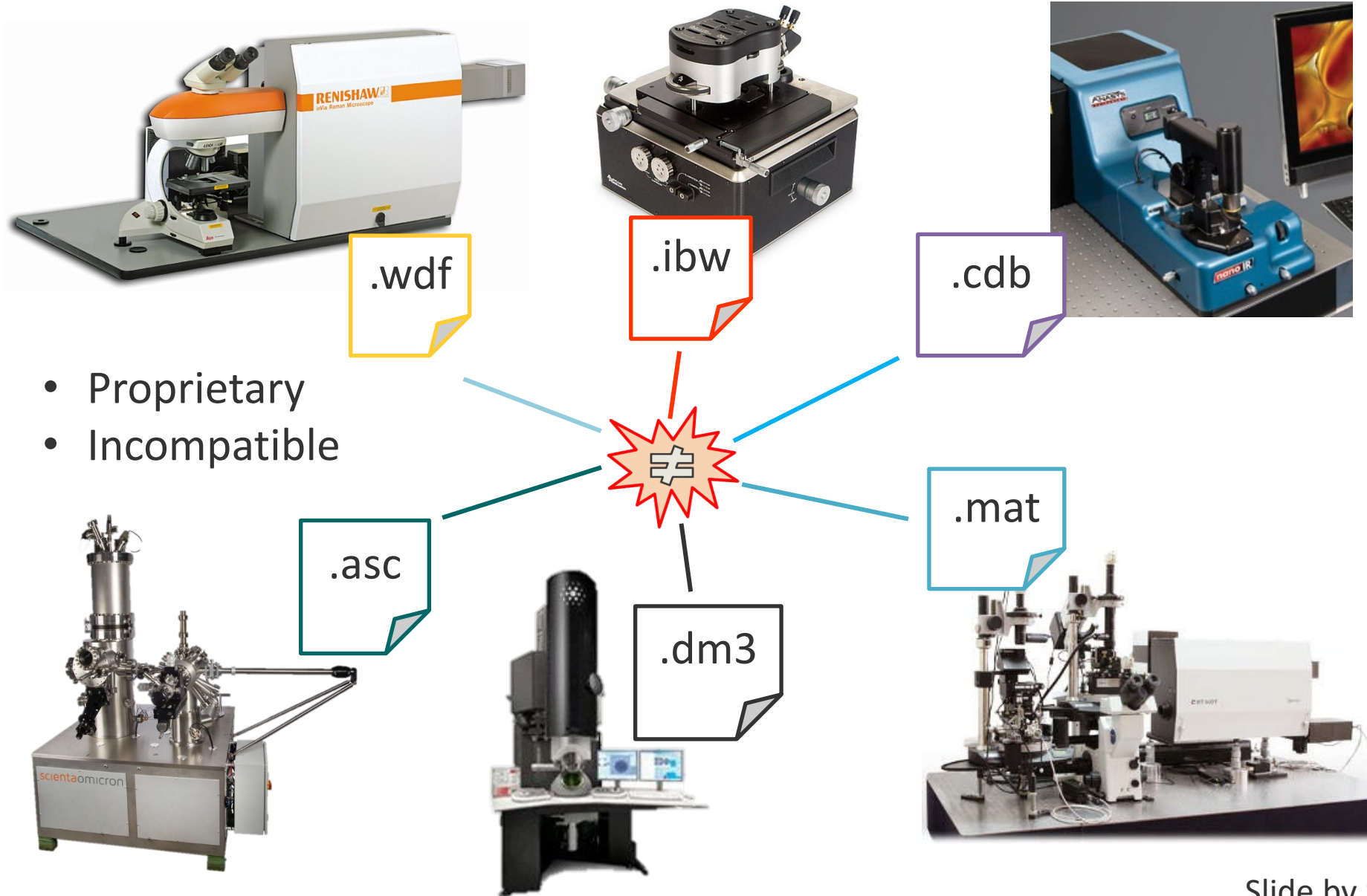


Scanning
Transmission
Electron
Microscope (STEM)



AFM with Raman
spectroscopy

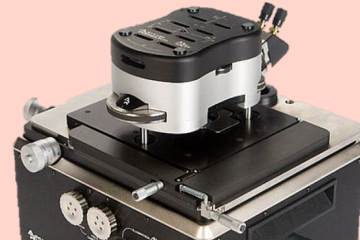
Multitude of File Formats



Disjointed communities....



- Clustering
- Fit spectra ...



- Filter Image
- Register Image ...

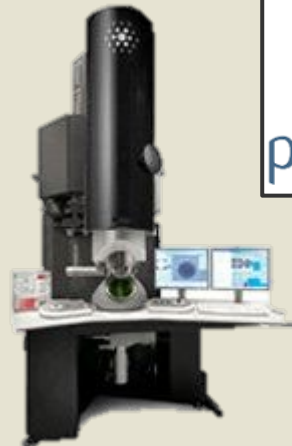


- Fit Spectra
- SVD Filtering ...

- FFT Filtering
- SVD Filtering ...



- FFT Filtering
- Classify Images ...

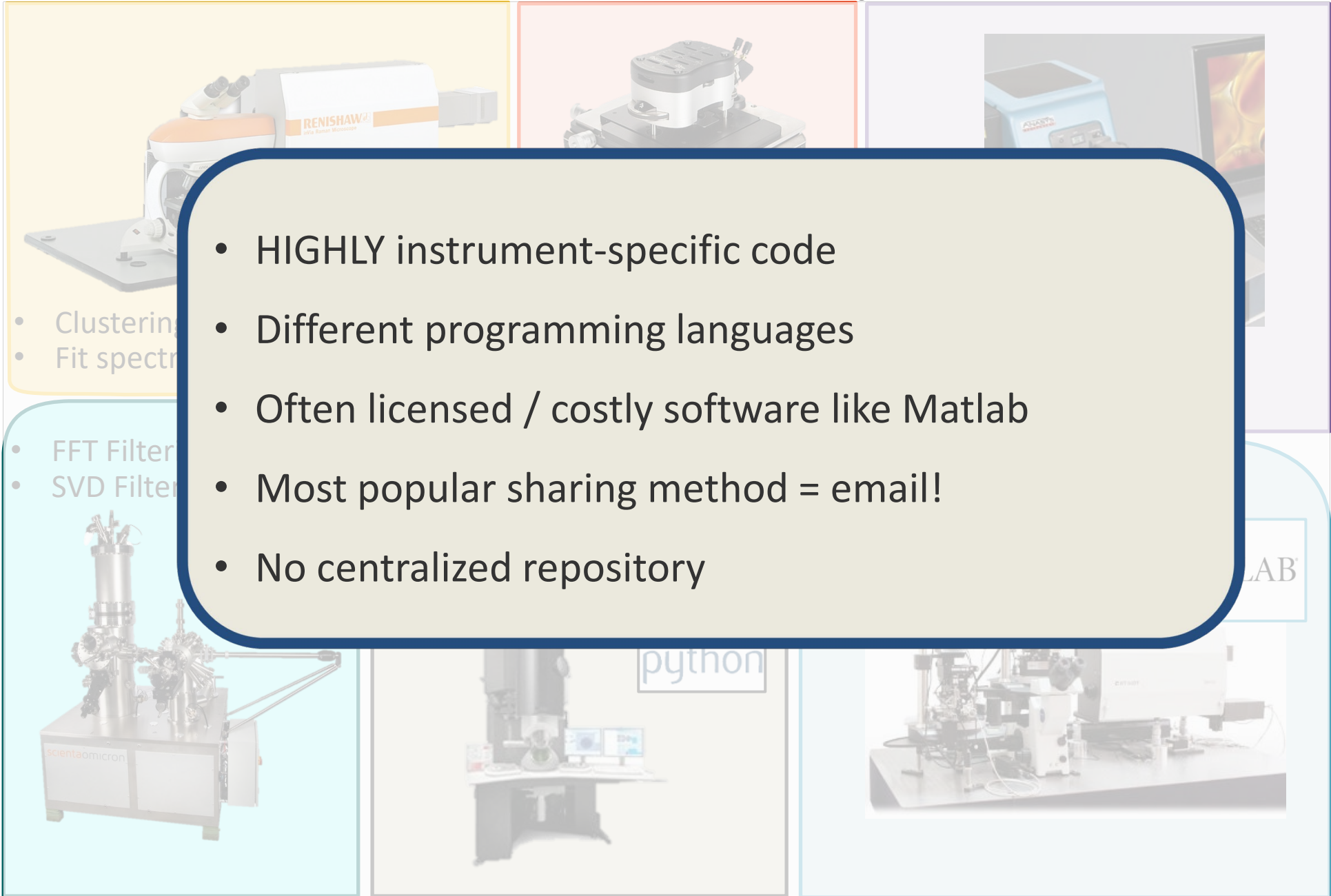


- Register Images
- Clustering



Slide by S. Somnath

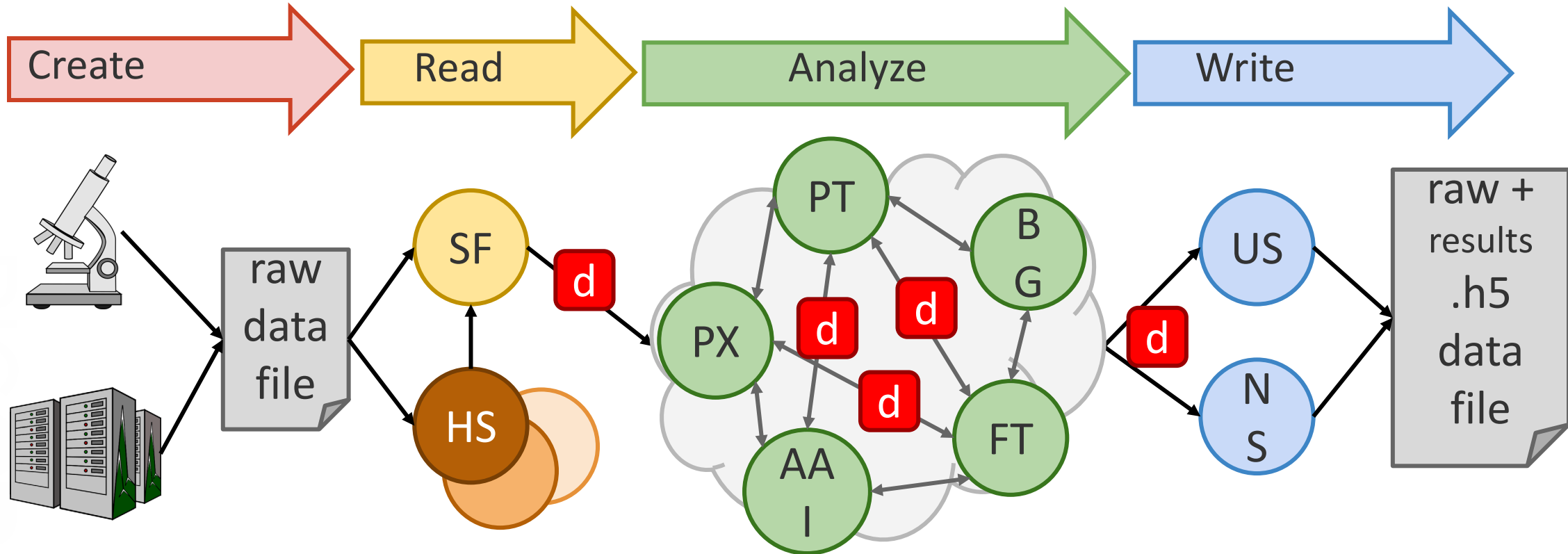
... cannot share code efficiently



- Clustering
- Fit spectra
- FFT Filter
- SVD Filter

- HIGHLY instrument-specific code
- Different programming languages
- Often licensed / costly software like Matlab
- Most popular sharing method = email!
- No centralized repository

Solutions: Integrated Ecosystems



Data from measurements or simulations are read into `sidpy.Dataset` (d) objects directly by **SciFiReaders** (SF). Data are processed using multiple science packages in the Pycroscopy ecosystem that interoperate via **Dataset** objects. **Dataset** objects are written to HDF5 files via **pyUSID** (US) or **pyNSID** (NS).

Solutions: Integrated Ecosystems



Instrument Tier



Automated, standardized,
modularized data acquisition



Instrument-agnostic, self-describing,
model in an open friendly file format



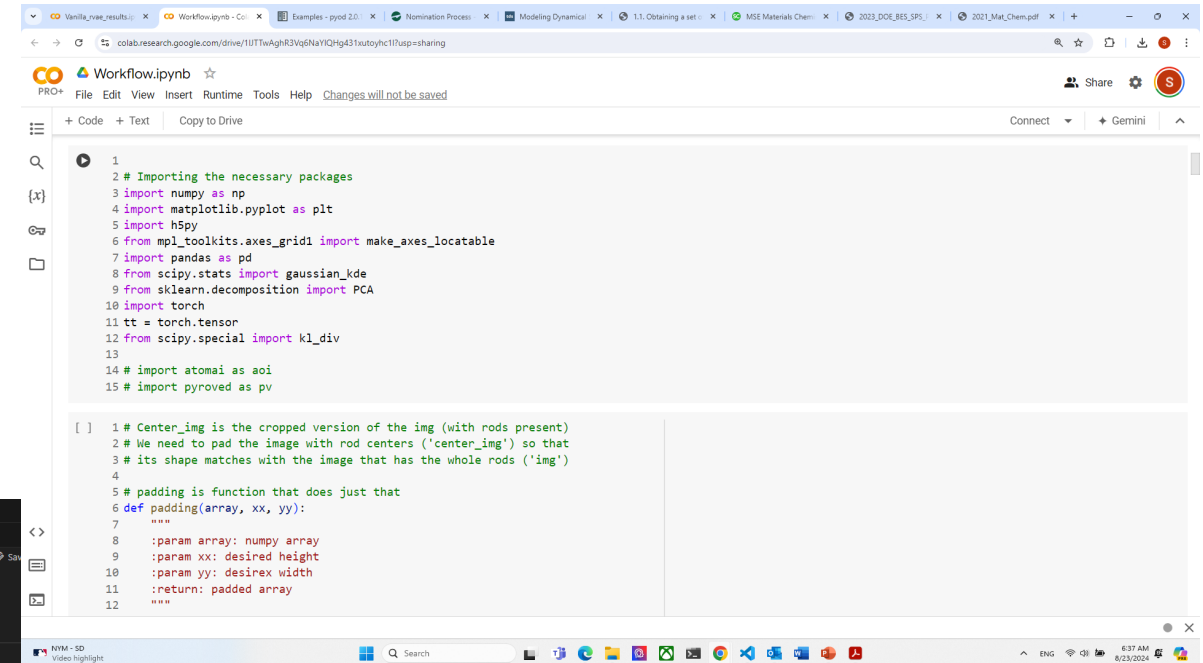
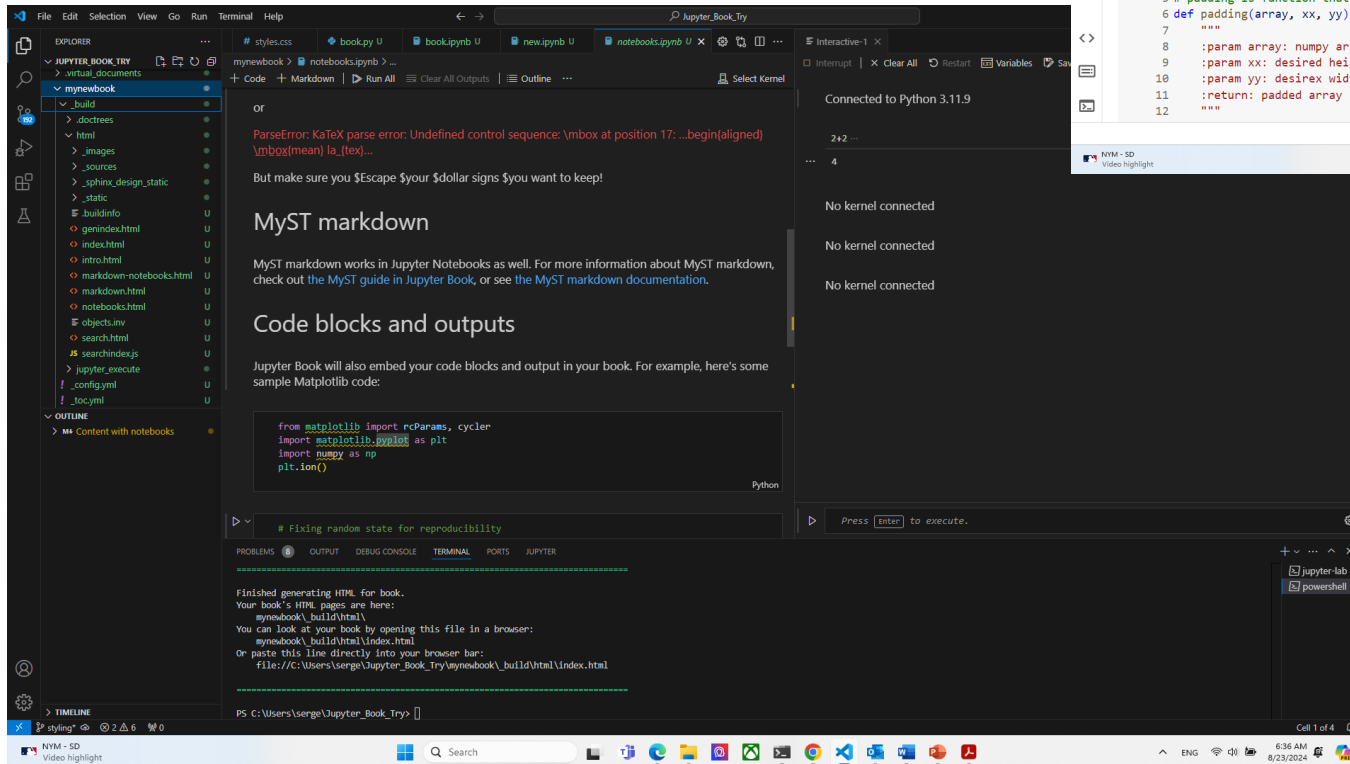
Centralized repository for data
processing, analysis



Interactive visualization + analysis +
storage on the cloud

How we can run code:

- Google Colabs
- AWS SageMaker notebooks
- IDE: Spyder, PyCharm, etc.
- Command line interface



Where would the code run?

- Cloud GPU
- Local computer
- Selected HPC (ISAAC for UTK)

What language do we use:

Main Python libraries we will use:

1. NumPy
2. Matplotlib
3. Scikit-learn
4. Keras

Other libraries we may use:

1. Seaborn
2. GPax
3. SciPy

- [This course](#)
- Read the docs
- Blogs (e.g. Medium)
- Packt, Manning, etc.
- Papers with codes
- GitHub repos

We will learn these as we need them!

We will also start to learn how to select sources

Code Repositories and Version Control

- Sharing scripts between users can be workable for immediate or short-term needs, but is not scalable nor lasting
- For reproducibility, it is better to have codes that reside in packages that are documented and well tested
- Most of you are familiar with python packages; but many are probably new to version control
- Version control systems such as git enable multiple people to work on a single software project at the same time to speed up development and ensure consistency
- Git is an open-source distributed version control system. It maintains a history of changes that have occurred in the project and allows for updates as well as reversions to older 'commits'.

How we can share code:

Git would take a significant amount of time to explain in detail. However, there are plenty of online tutorials, e.g.

<https://www.atlassian.com/git/tutorials>

Net Ninja, Git and GitHub tutorial for beginners



YouTube player interface showing a video titled "Git & GitHub Tutorial for Beginners #1 - Why Use Git?" by Net Ninja. The video is part of a series of 12 videos. The player shows the video progress bar at 0:03 / 5:45. The video content displays the "THENETNINJA" logo.

Git & GitHub Tutorial for Beginners #1 - Why Use Git?

Net Ninja 1.54M subscribers

542K views 7 years ago

Donate via PayPal - <https://www.paypal.me/thenetninja>

Series: Git & GitHub Tutorial for Beginners

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- 4. Git & GitHub Tutorial for Beginners #4 - Creating a...
- 5. Git & GitHub Tutorial for Beginners #5 - Staging files
- 6. Git & GitHub Tutorial for Beginners #6 - Making...
- 7. Git & GitHub Tutorial for Beginners #7 - Undoing Things
- 8. Git & GitHub Tutorial for Beginners #8 - Branches
- 9. Git & GitHub Tutorial for Beginners #9 - Merging...

Download the eBook: 50 KPI Checklists to Drive Your Company's Success

https://www.youtube.com/watch?v=3RjQznt-8kE&list=PL4cUxeGkcC9goXbgTDQ0n_4TBzOO0ocPR

Get data – from publications

- Printed matter
- Pdf files with figures and tables
- Deposited data files
- (Rarely) workflows

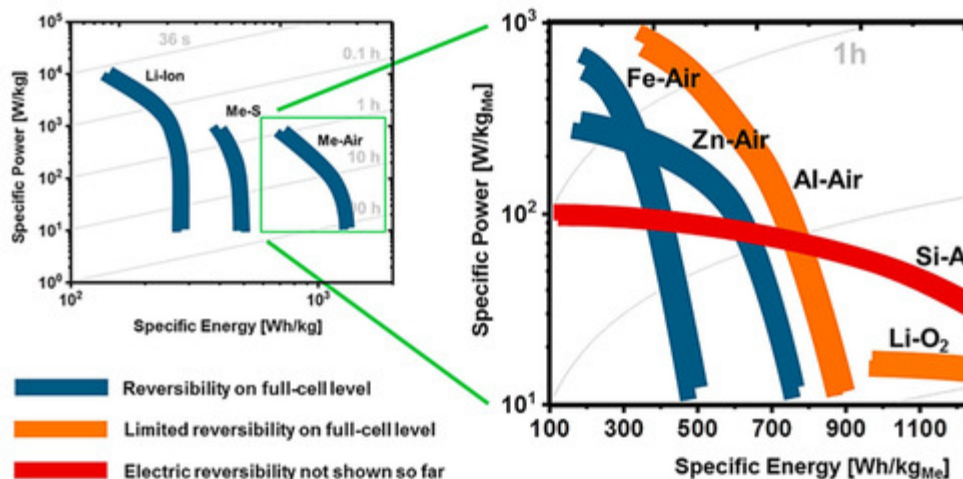
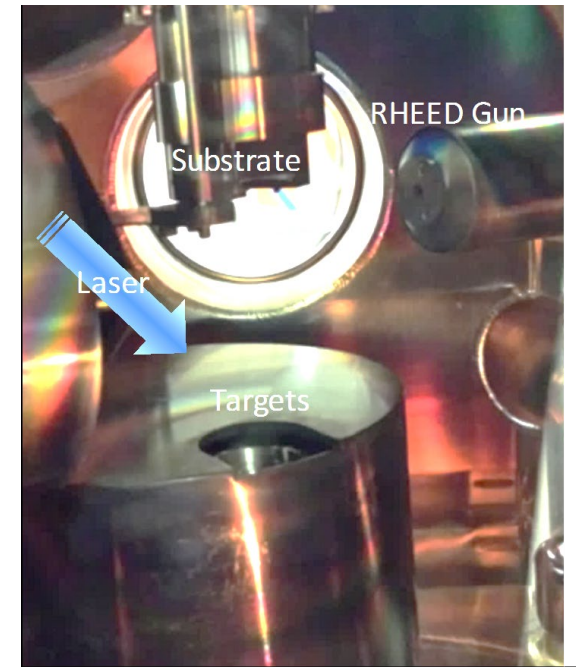
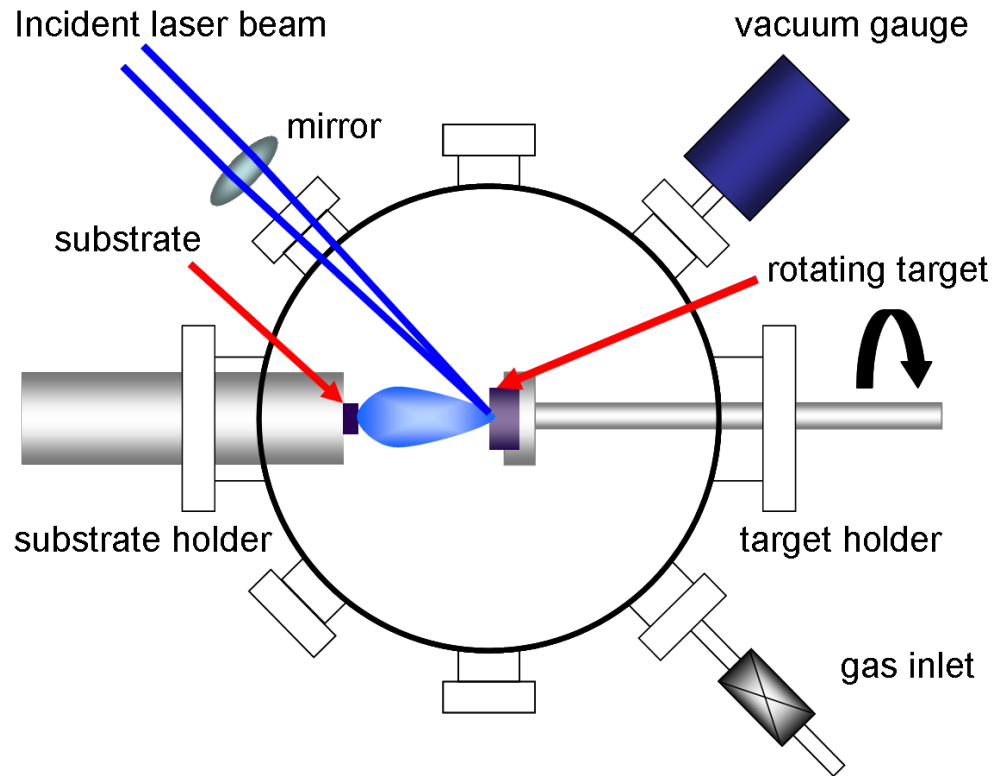


Table 5. The recent experimental results of different MABs, adapted from [58].

MAB	Discharge Product	Experiment Specific Energy (Wh kg ⁻¹)	Condition	Reversibility Cycles	Voltage (V) Ref.
Fe/O ₂	Fe(OH) ₂	453 Wh/kg _{Fe}	[b, c, d, e]	3500 [b, d]	1.28
Zn/O ₂	ZnO	>700 Wh/kg _{Zn}	[a, c, d]	>75 [a, c]	1.65
K/O ₂	KO ₂	~19,500 Wh/kg _{Carbon}	[a, c, d]	>200 [a, c]	2.48
Na/O ₂	Na ₂ O ₂	~18,300 Wh/kg _{Carbon}	[a, c, d]	>20 [a, c]	2.33
Mg/O ₂	Mg(OH) ₂	~2750 Wh/kg _{Oxide}	[a, c, d, f]	<10 [a, c, d]	2.27
Si/O ₂	Si(OH) ₄	~1600 Wh/kg _{Si}	[a, c, d]	Not yet	2.09
Al/O ₂	Al(OH) ₃	~2300 Wh/kg _{Al}	[a, c, d]	Limited	2.71
Li/O ₂	Li ₂ O ₂	>11,050 Wh/kg _{Carbon}	[a, c, d]	>250 [a, c]	2.96
	Li ₂ O				2.91

Conditions: a is anode sheet/foil, b is porous/particulate anode, c is full-cell measurements, d is 100% deep discharge, e is repeated charge/discharge, and f is elevated temperature.

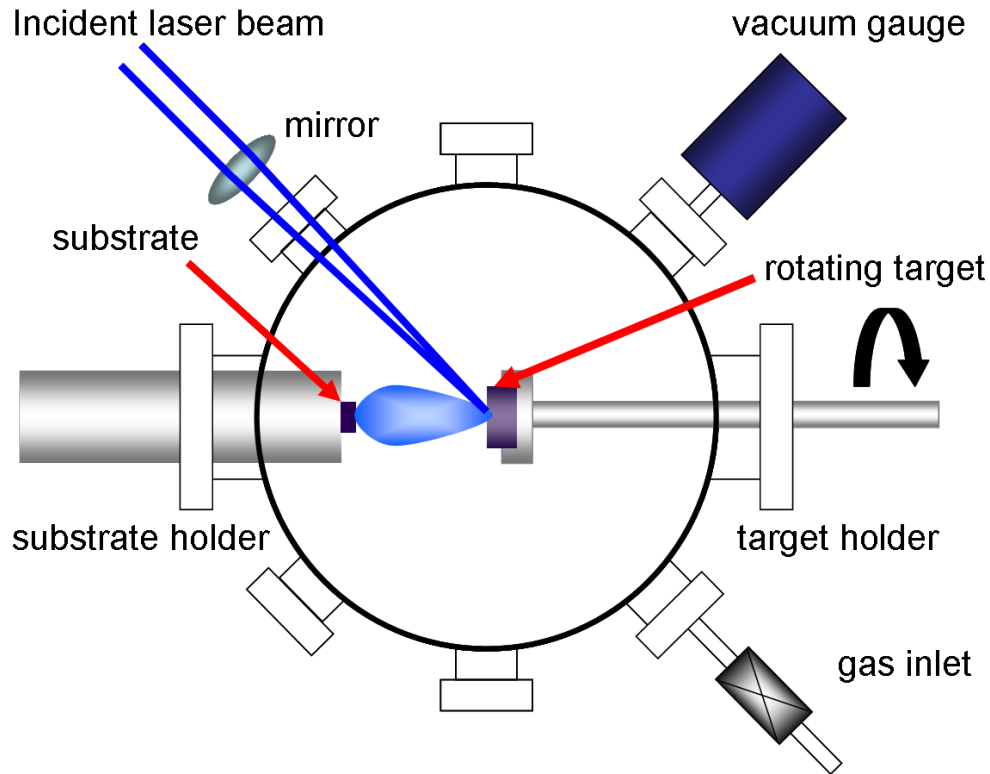
Pulsed Laser Deposition



- PLD benefits: great films, ease of setup, wide variety, unit-cell precision in growth
- Downside: difficult to correlate growth parameter with film properties, largely trial and error approach



Pulsed Laser Deposition



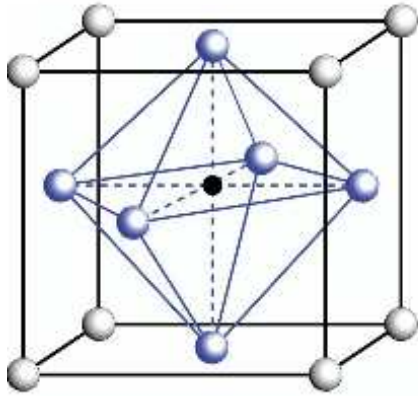
Parameters to vary:

1. Substrate (compound, orientation)
2. Film (compound)
3. Growth temperature
4. Growth environment (e.g., pO_2)
5. Laser Fluence and repetition rate
6. Target-substrate Distance
7. Heterostructures (film electrodes, buffer layers, etc.)
8. Post-annealing
9. Cooling Rate

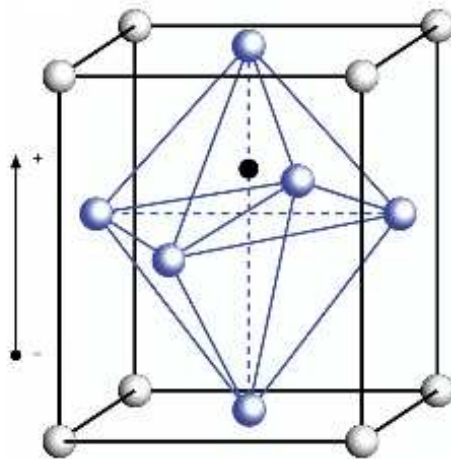
- Researchers will grow films by varying parameters, achieving different **functional property** results. These could include film roughness, stoichiometry, and specific properties, such as capacitance, remnant polarization, Curie temperature, etc.
- Iterative procedure, since PLD conditions are generally not transferable.

Ferroelectric Materials

High T

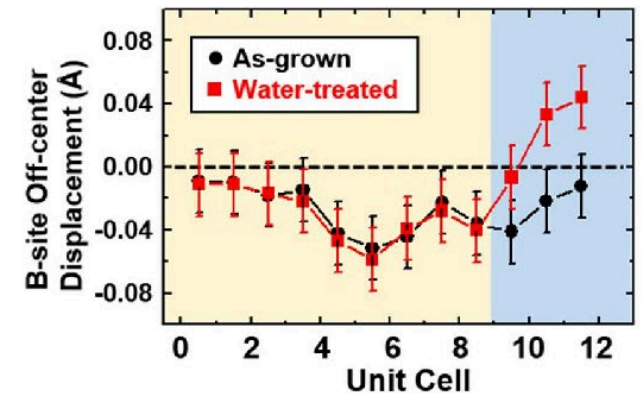
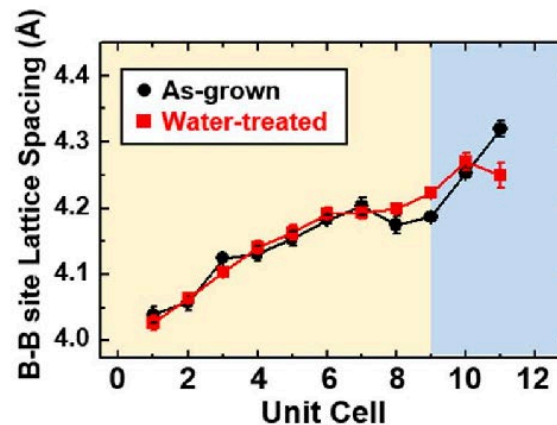
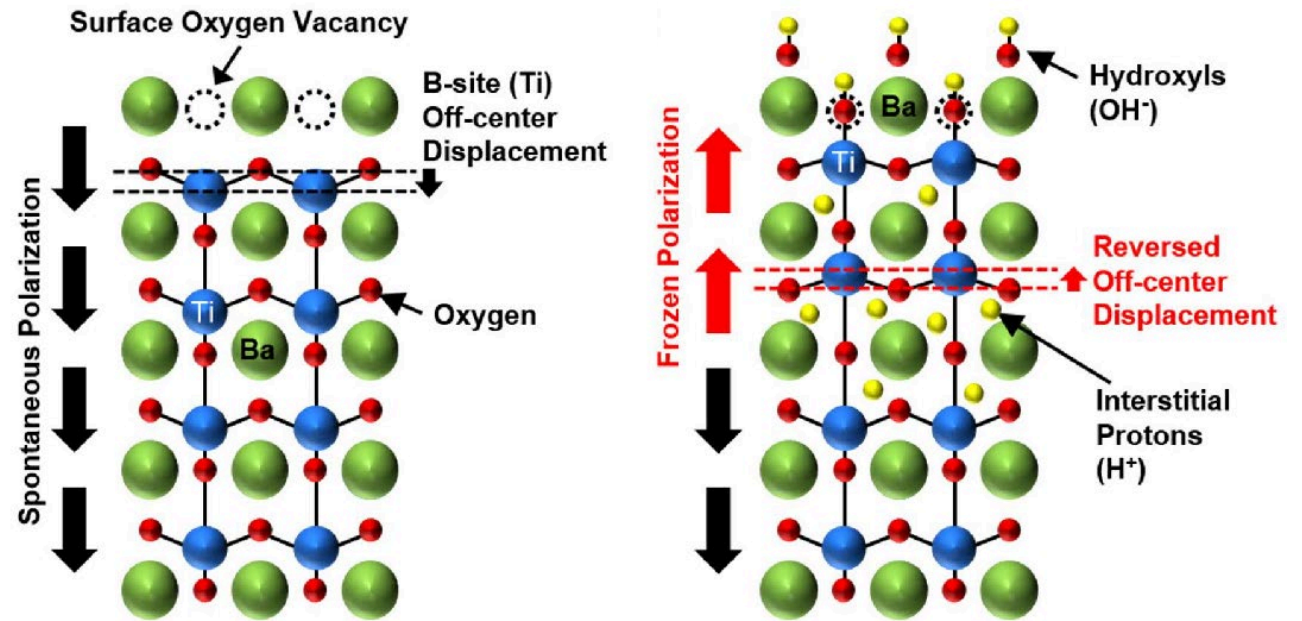


Low T

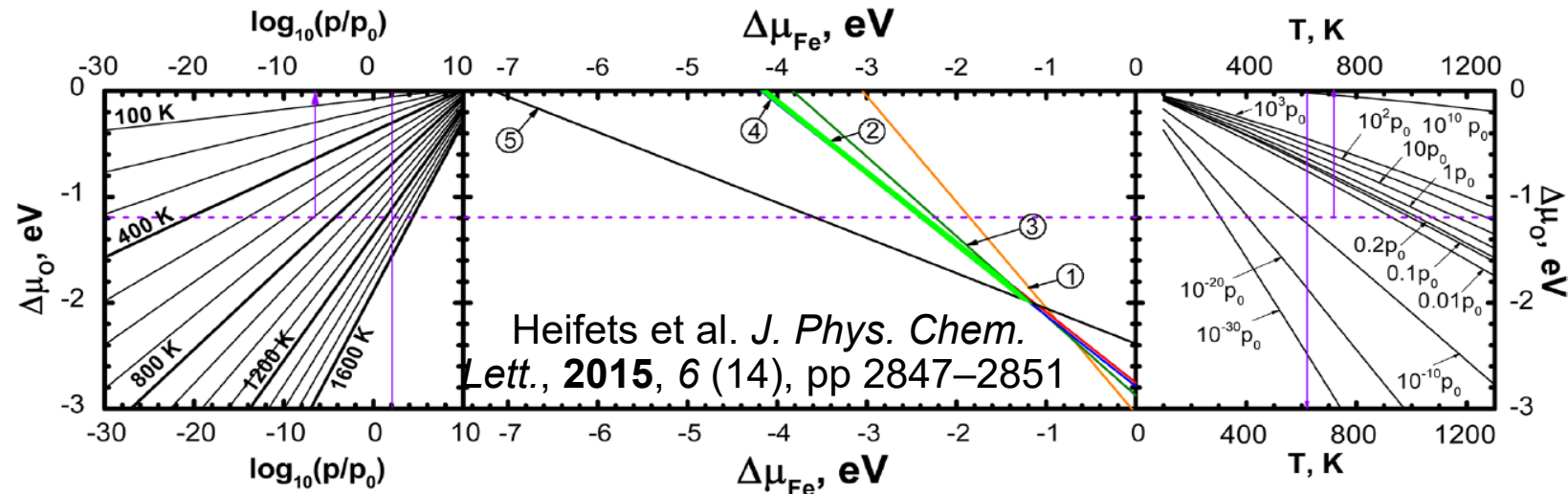


- Can store information
- Sensitive to charge
- Sensitive to strain

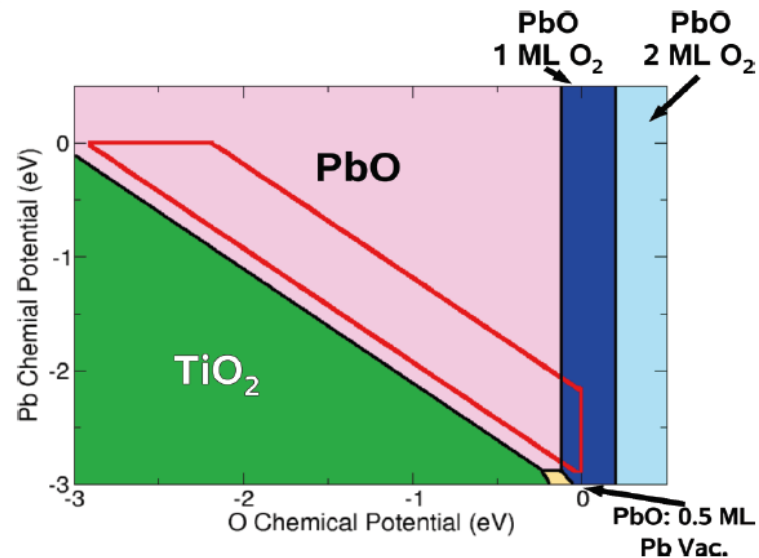
Surfaces and interfaces matter!



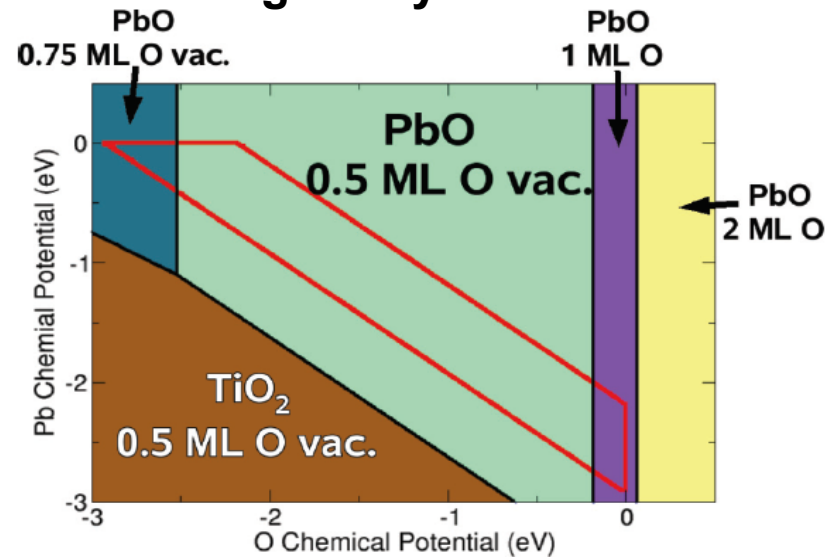
Can theory help?



Positively Poled



Negatively Poled



Garrity et al. PRB 88, 045401 (2013)

Practical Considerations

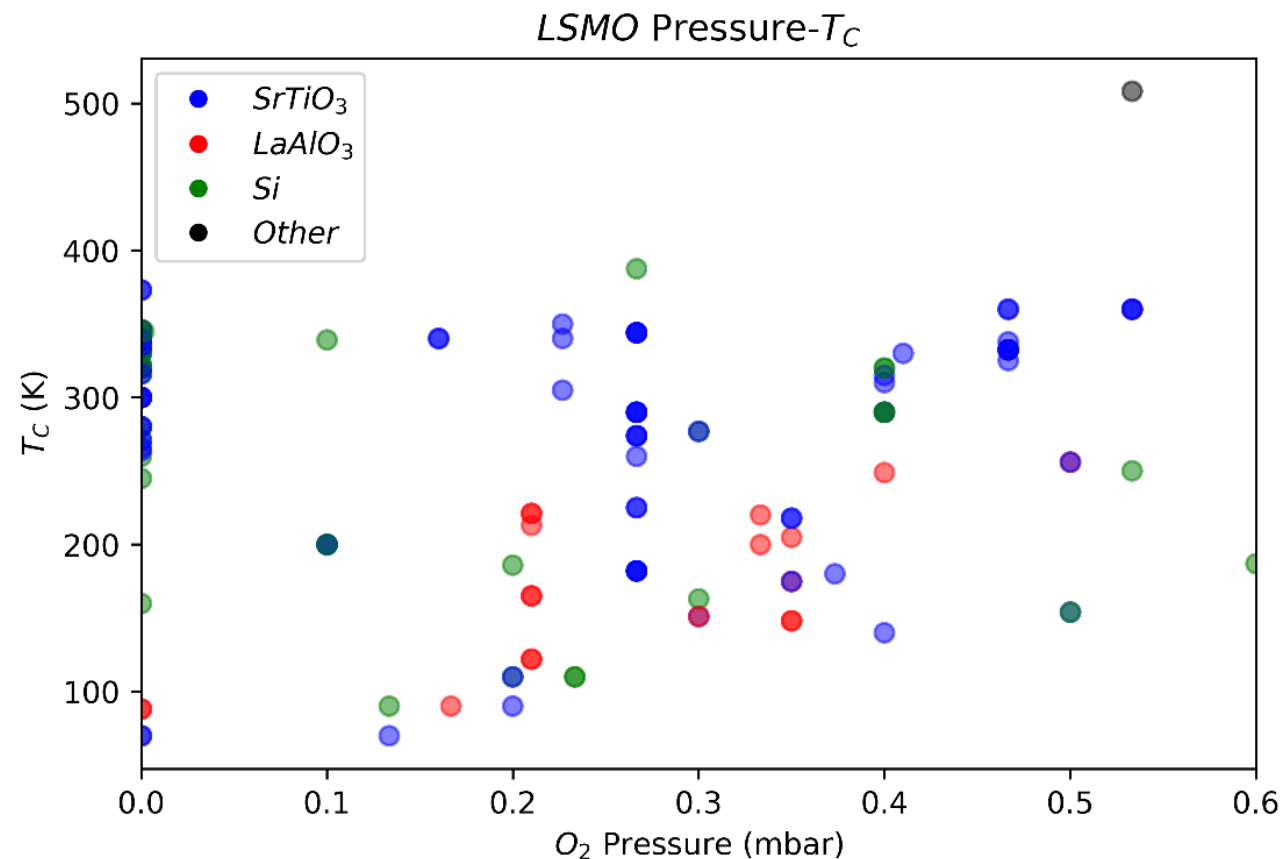
1. DFT is not always accurate, often doesn't account for defects.
2. Kinetic factors during growth.
3. Ideal stoichiometry not always correlated to best properties.

So, we need to know what conditions to use practically...

Slide courtesy of R. Vasudevan

We can mine publications for data. But...

- We get coercive temperature dependent on growth temperature and partial oxygen pressure.
- What's next?



RESEARCH ARTICLE | MARCH 20 2018

Data mining for better material synthesis: The case of pulsed laser deposition of complex oxides ✓

Steven R. Young ; Artem Maksov; Maxim Ziatdinov; Ye Cao; Matthew Burch; Janakiraman Balachandran; Linglong Li; Suhas Somnath; Robert M. Patton; Sergei V. Kalinin ; Rama K. Vasudevan

Check for updates

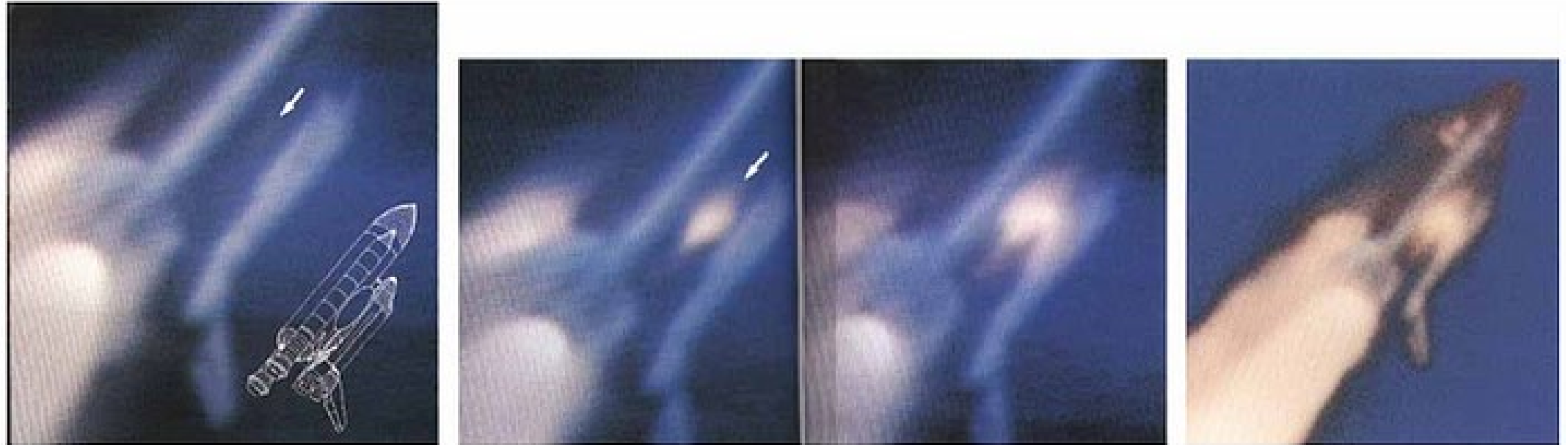
+ Author & Article Information

J. Appl. Phys. 123, 115303 (2018)

<https://doi.org/10.1063/1.5009942> Article history

<https://pubs.aip.org/aip/jap/article/123/11/115303/155202/Data-mining-for-better-material-synthesis-The-case>

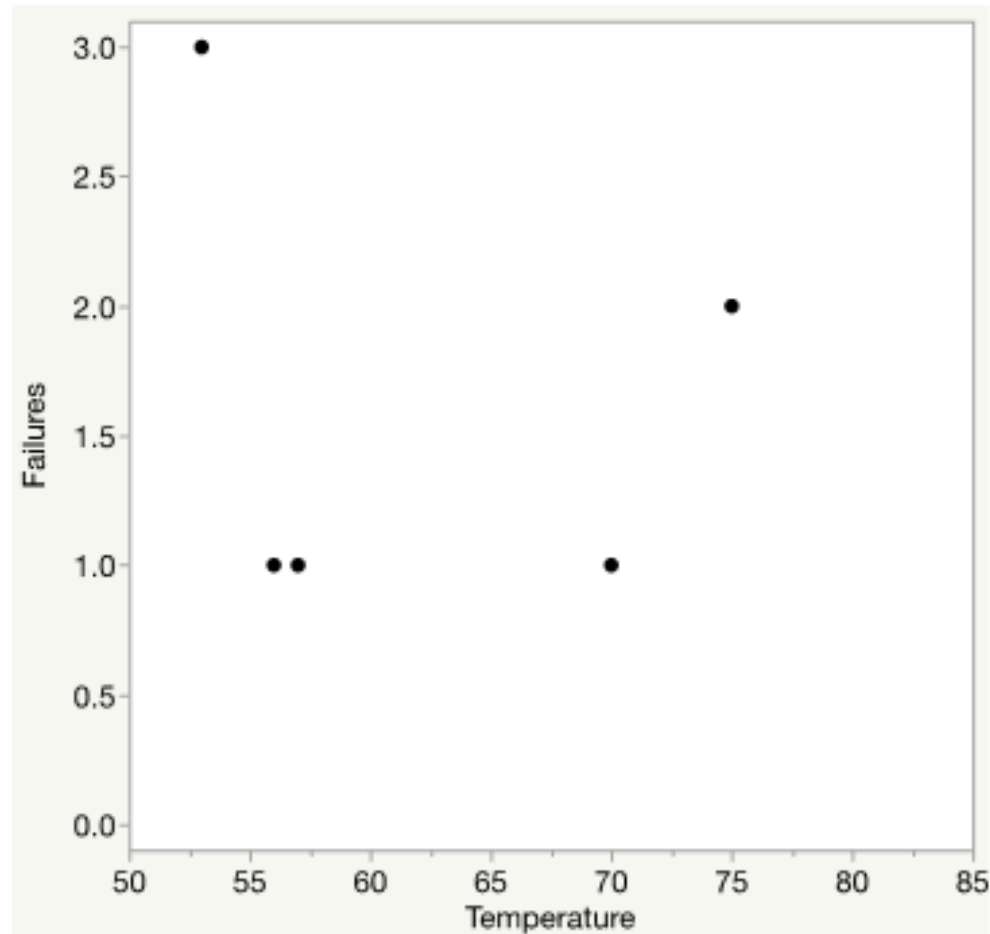
Challenger disaster



<https://medium.com/habits-for-success/the-challenger-disaster-a-lesson-in-the-power-of-data-analysis-and-visualization-398d2ac8f59b>

<https://www.youtube.com/watch?v=hOE1t1MOuYX4>

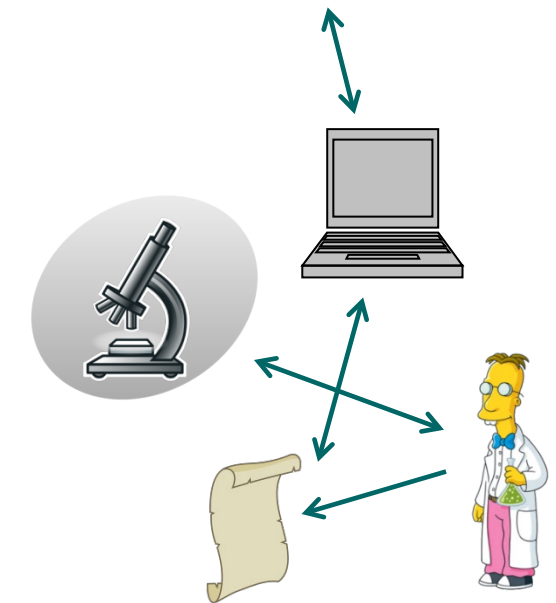
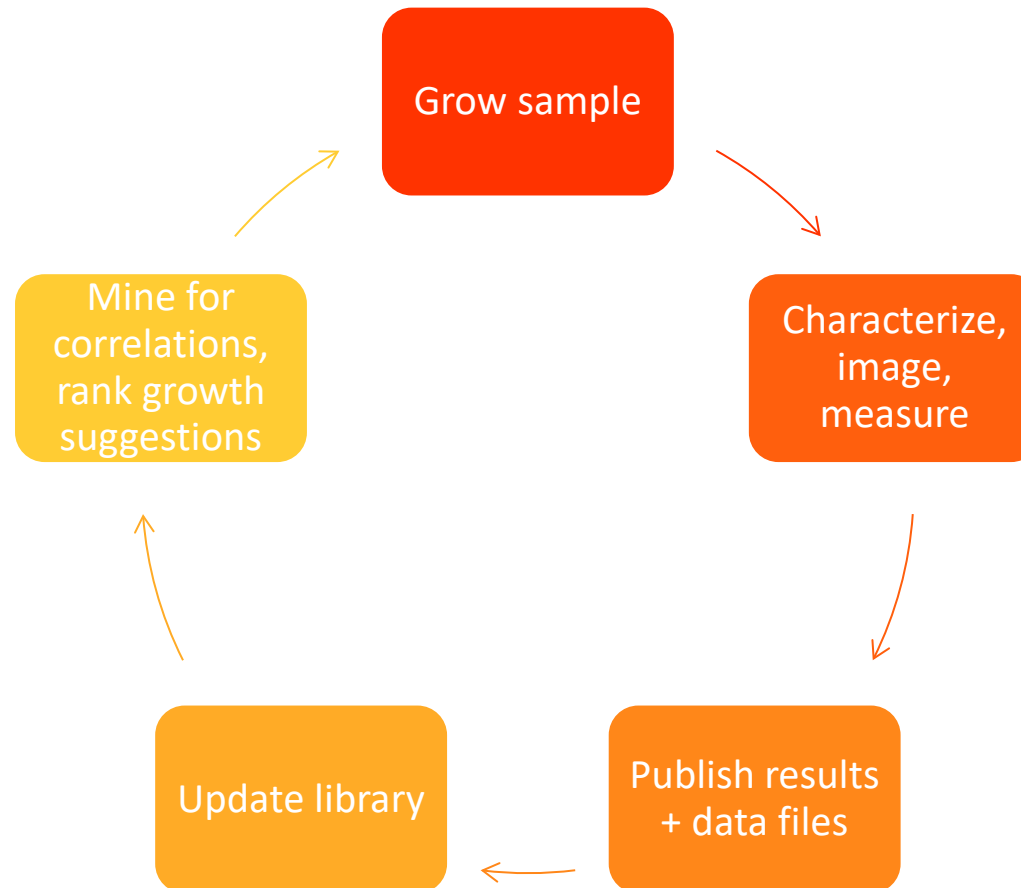
Challenger disaster



https://www.researchgate.net/figure/Challenger-Data-Number-of-O-Ring-Failures-Launch-Versus-TemperatureLeft-Panel-Five_fig2_344257416

Conclusion: requires community involvement

- Relevant sample preparation conditions
- Unique sample identifier
- Measured properties in appropriate format



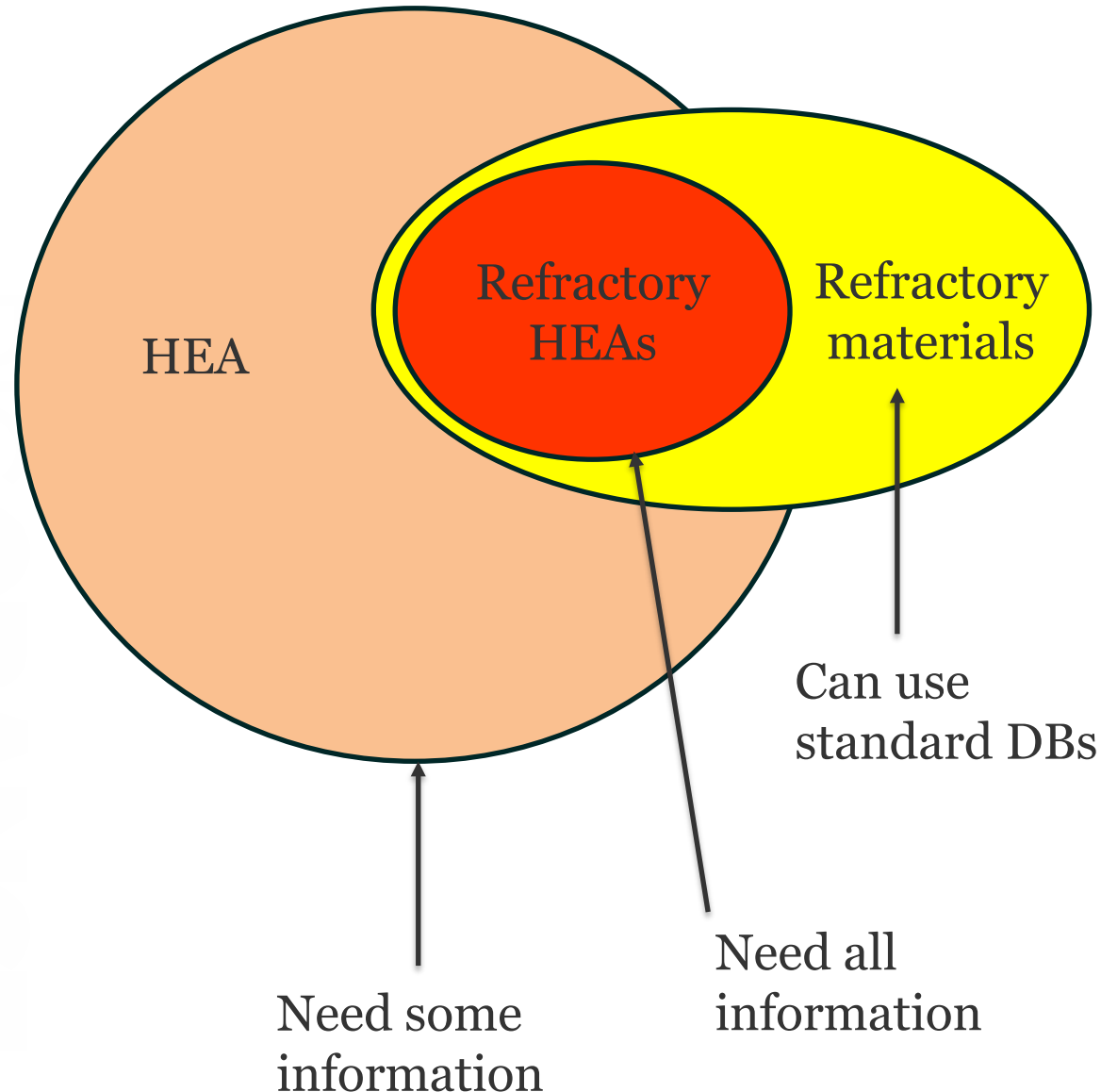
Get data – data bases!

- **On-line data bases**

- Materials Project, <https://next-gen.materialsproject.org/>
- Materials Data Facility: <https://materialsdatafacility.org/>
- Papers with code: <https://paperswithcode.com/datasets>
- Standard data sets for molecular properties, e.g. <http://quantum-machine.org/datasets/>
- Integrators, e.g.: <https://github.com/sedaoturak/data-resources-for-materials-science>

- **Enterprise data bases**

We Start with Data



To run active learning experimental campaign for refractory HEA, we need to understand how property emerge in HE systems and refractory systems

For refractory HEA:

- Ideally collect all data out there
- Need targeted literature mining

For HEAs in general

- Find data bases out there
- Literature mining if simple

For refractory materials in general

- Use existing DBs

Static schemas are a good start, but we need much more (synthesis, workflows, heuristics)!

Get data – LLM Analysis!

- **Use ChatGPT to:**

- Analyze collection of abstracts
- Mine the paper for data
- Summarize paper
- Reproduce code in the paper



**Homework
assignment 2
(part 2)**

- **(advanced) Use LangChain to train ChatGPT on:**

- Multiple papers
- Finetune on arxiv, chemrxiv, etc

- **Expect more changes in the near future!**

Homework 2:

- Use ChatGPT/NotebookLM for paper analysis
- Also possible start for final project!

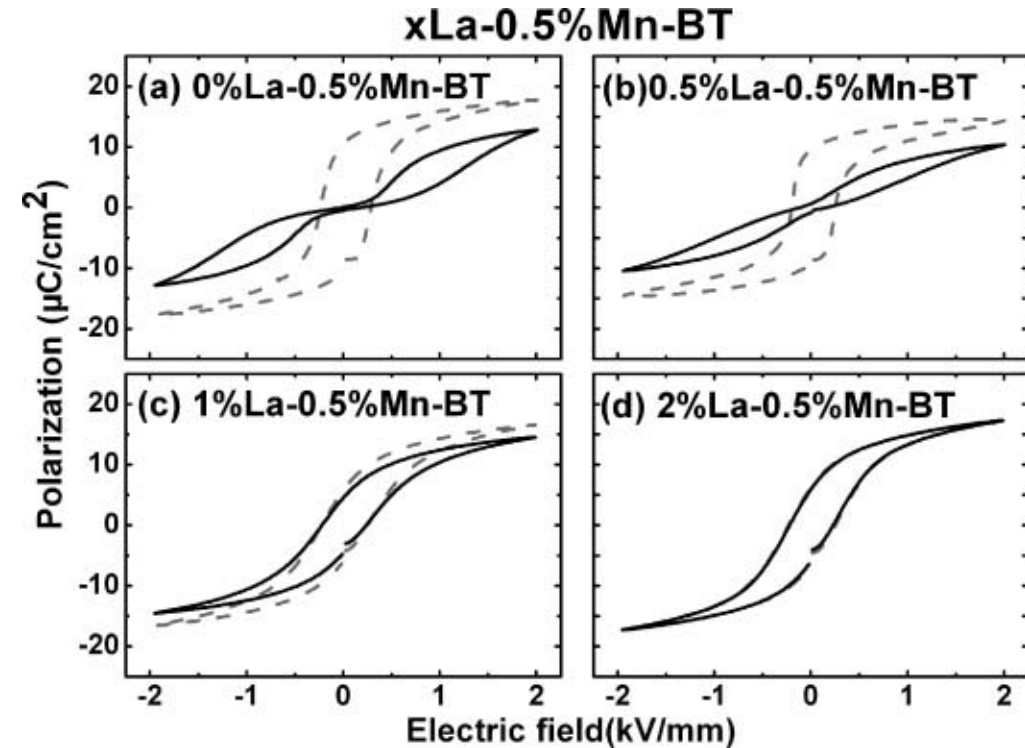
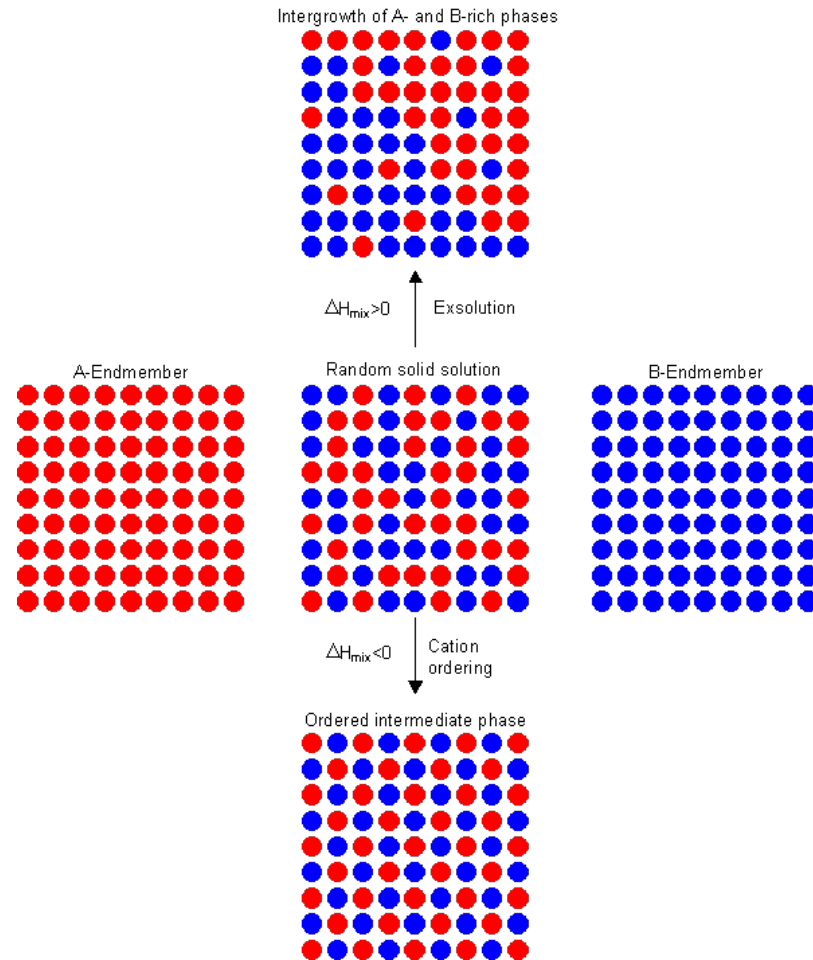
Hackathon: October 21-22, <https://llmhackathon.github.io/>

Extra slides

ChatGPT Example

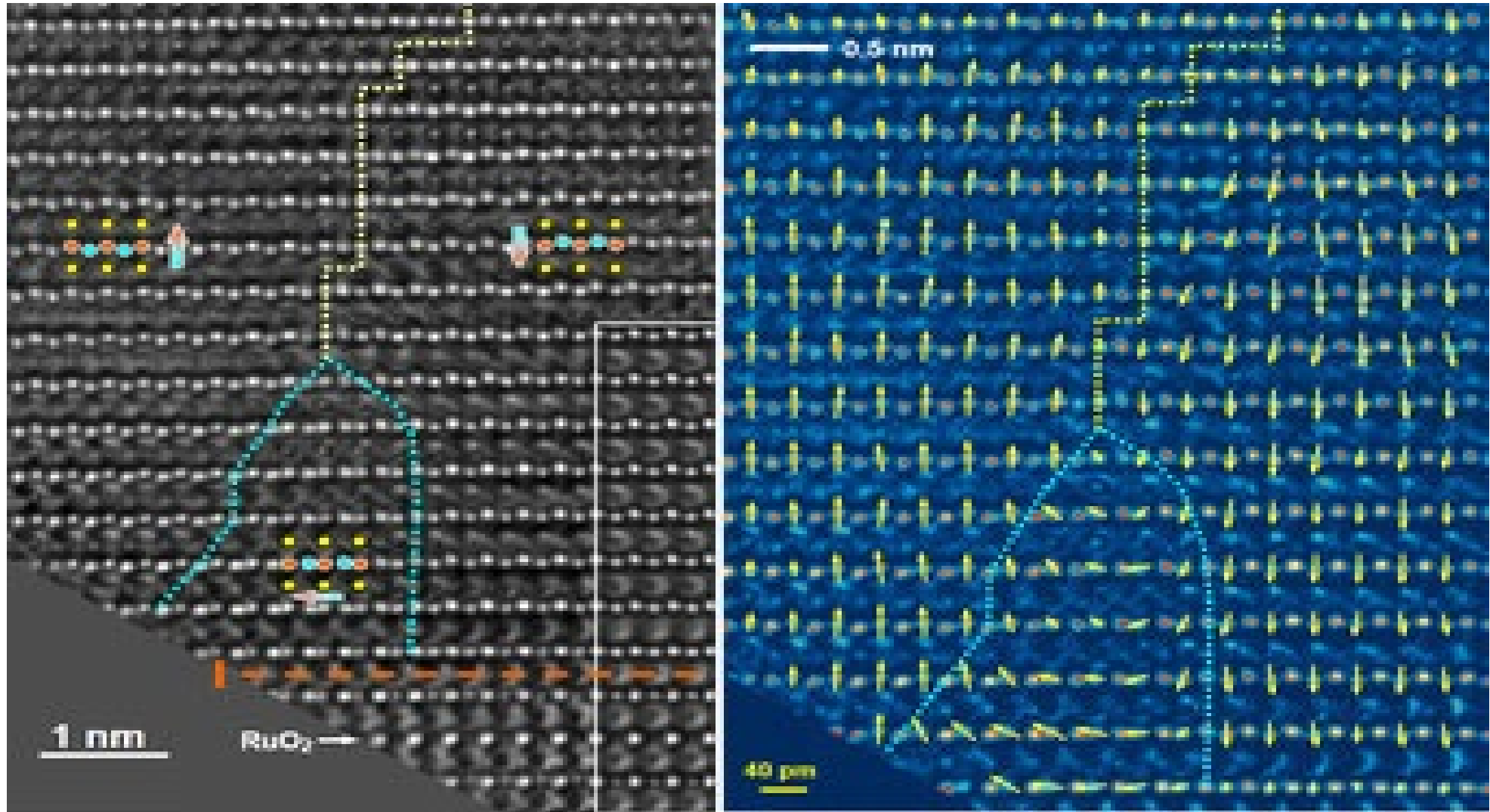
1. "Let's analyze scientific papers. I will upload three papers, and you will extract information about materials. Ok?"
2. "This is paper 1"
3. "This is paper 2"
4. "This is paper 3"
5. "Make a table of the compounds studied in these three papers, including name, formula, and lattice parameters."
6. "What parameters are known for all studies compounds?"
7. "Has these compounds been studied by XRay scattering?"
8. "Can you extract positions of XRay peaks?"
9. "Make a list of prompts that I made"

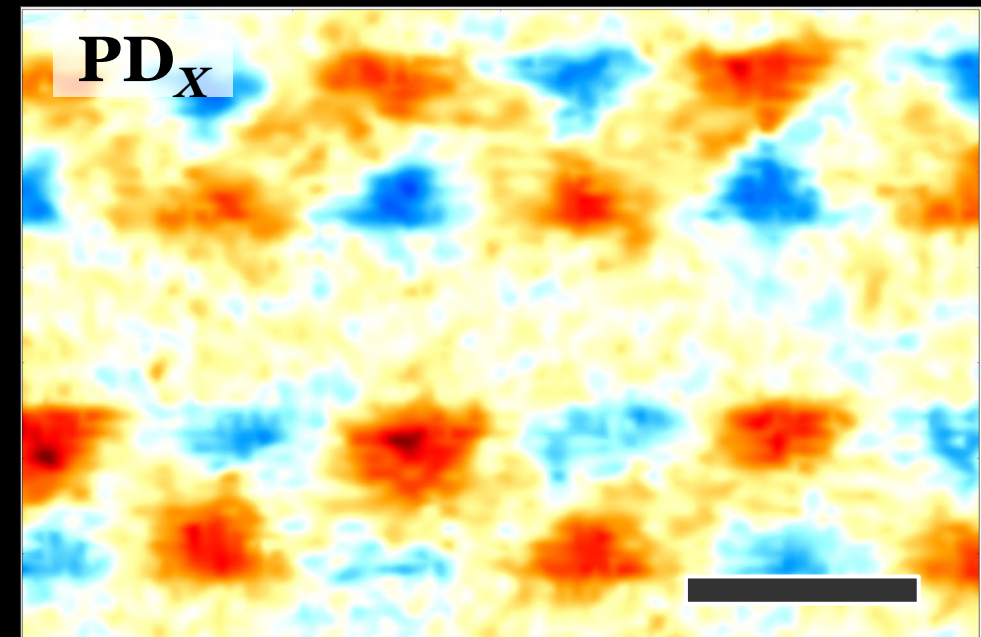
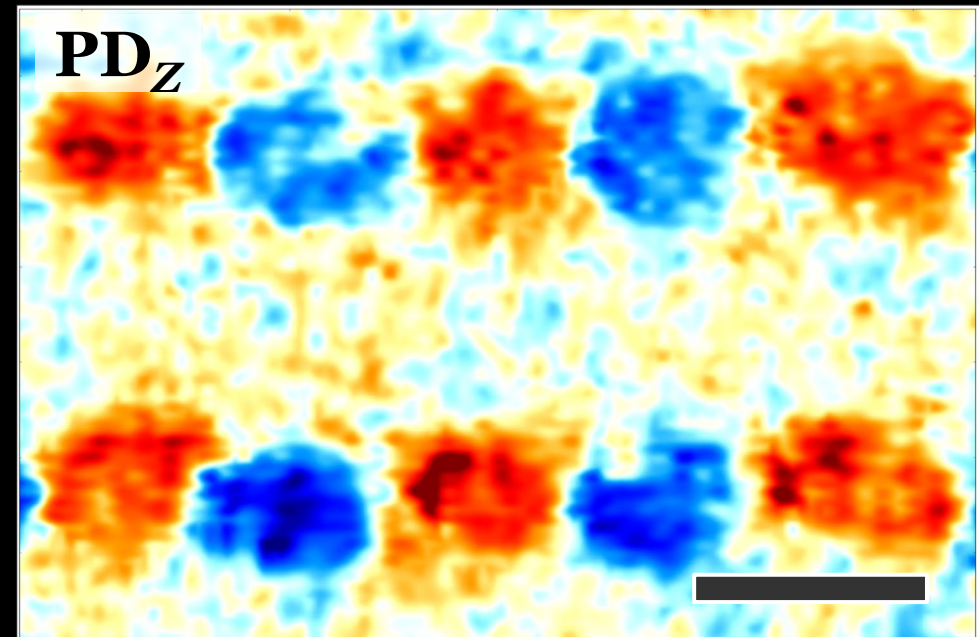
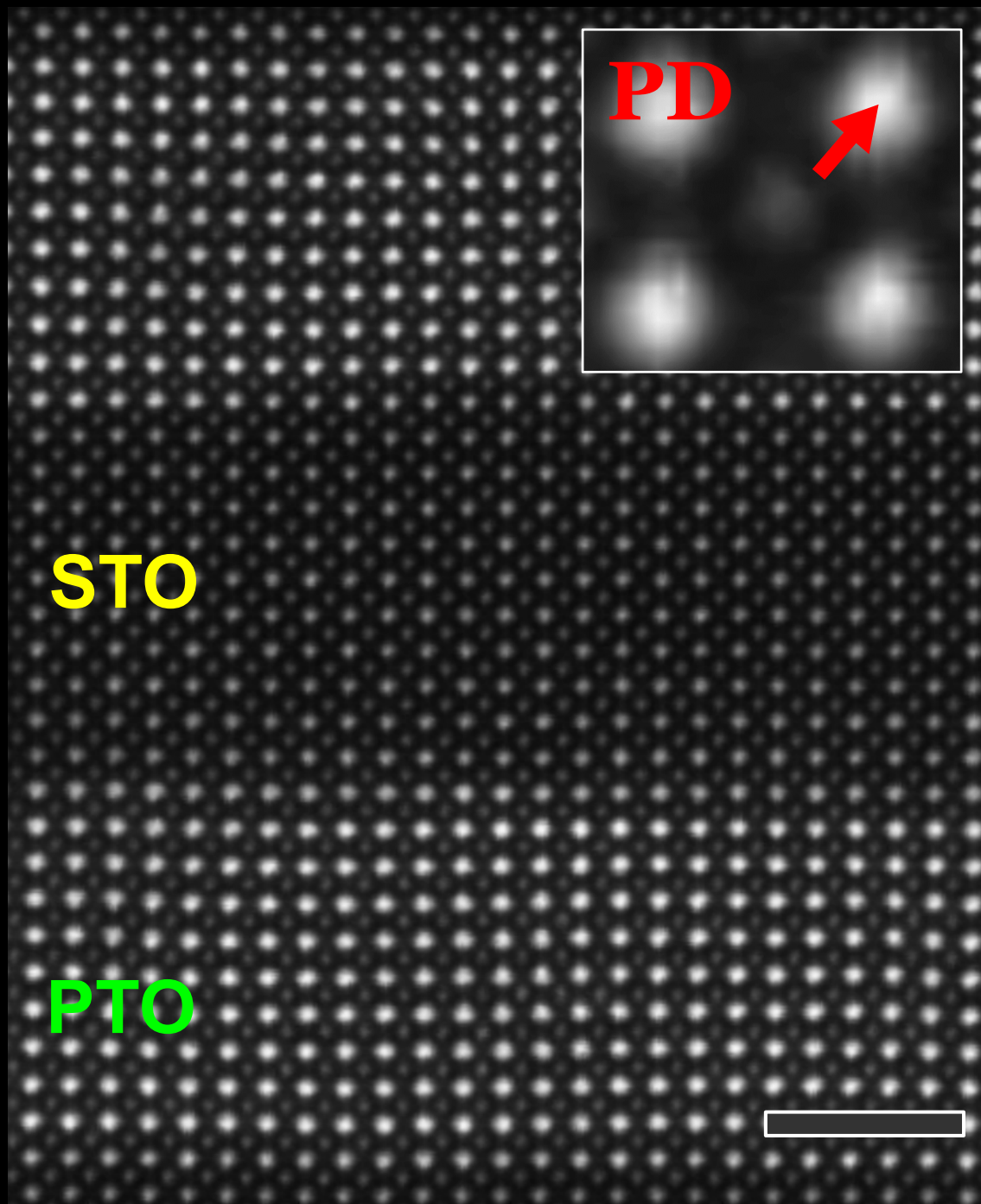
Cause and effect in doped ferroelectrics



- We generally assume that cationic order is frozen at the state of material formation, and then polarization field evolves to accommodate charged dopants.
- However, ions can move to compensate polarization – segregation at the domain walls, memory effects, etc.

Direct Observation of Atomic Structure

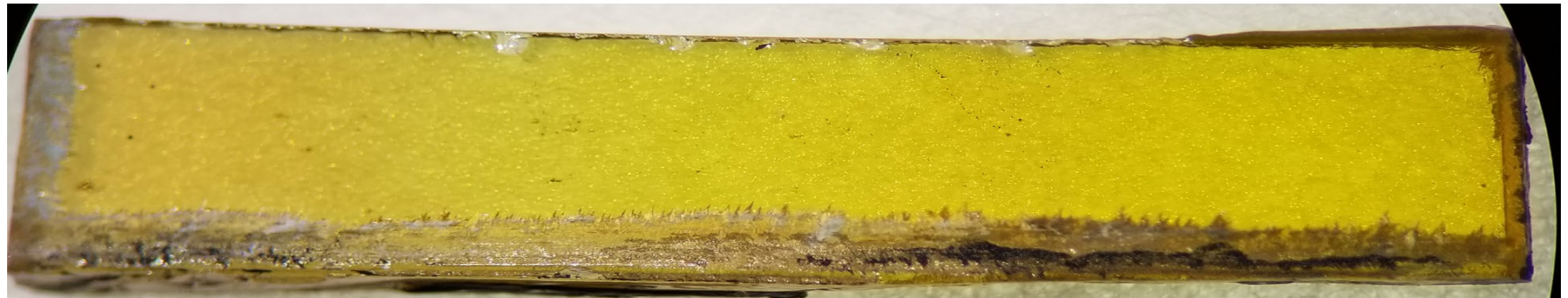




Cause and effect in doped ferroelectrics

For real material, can we establish what is the cause and what is the effect:

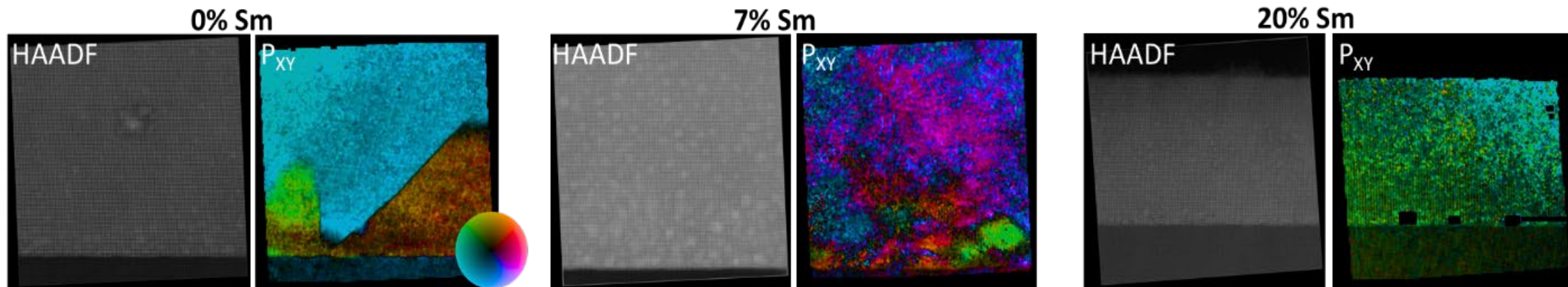
- Does polarization align to the cationic disorder
- Or does polarization instability drive cationic disorder?



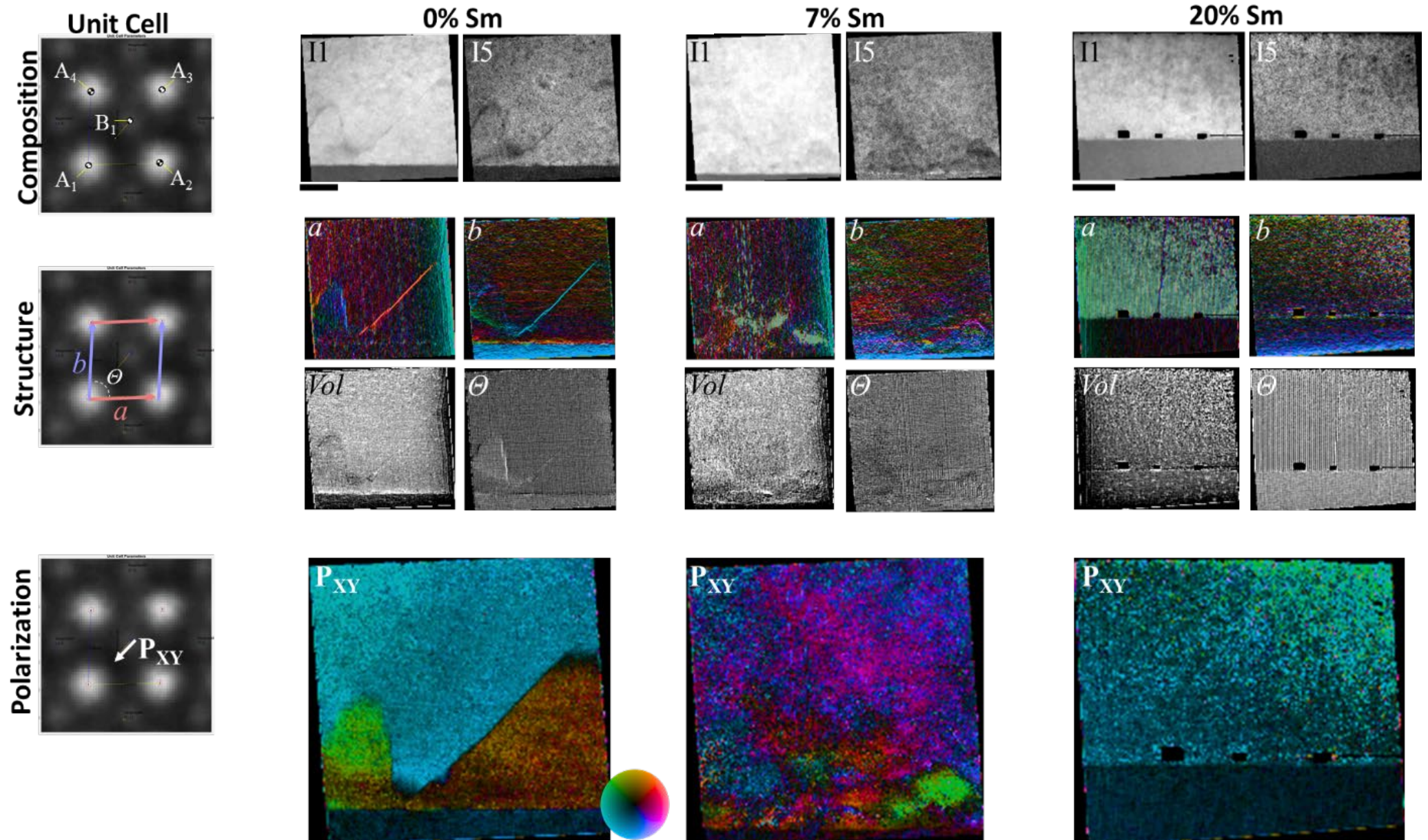
20%Sm BiFeO₃

Linear est. 7%Sm BiFeO₃

BiFeO₃



Building descriptor banks



Data-based causal analysis

Step 1: Identify conditional independencies between variables.

Step 2: Gradually remove edges in a fully connected graph to create a skeleton that represents potential causal relationships.

Step 3: Orient edges based on these independencies to form a DAG

```
for entry in SBFOdata:
    print(entry.keys())
```

```
def extract_physical_values(entry):
    return {
        'Alkali_Cations': entry['I1'].flatten(),
        'Transition_Metal_Cations': entry['I5'].flatten(),
        'Lattice_Parameter': entry['a'].flatten(),
        'Composition': entry['Composition'],
        'Unit_Cell_Angle': entry['alpha'].flatten(),
        'Volume': entry['Vol'].flatten(),
        'In_Plane_Polarization': entry['Pxy'][0].flatten(),
    }
```

```
# PC discovery without LLM assist
```

```
pc = PC(variant='stable')
```

```
pc.learn(scaled_data)
```

```
# Create the inverse mapping from indices to variable names
```

```
inverse_var_map = {v: k for k, v in all_vars.items() }
```

```
# Replace the indices with variable names in the DAG matrix
```

```
labels = [inverse_var_map[i] for i in
```

```
range(pc.causal_matrix.shape[0])]
```

```
pc_dag_named = pd.DataFrame(pc.causal_matrix, index=labels,
                             columns=labels)
```

PC-Discovered DAG Matrix without LLM assist

In_Plane_Polarization	1	0	1	1	0	1	0
Volume	1	0	1	1	0	0	0
Unit_Cell_Angle	1	0	1	1	0	1	0
Composition	1	0	0	0	0	0	0
Lattice_Parameter	1	0	0	1	0	0	0
Transition_Metal_Cations	1	0	0	1	0	1	1
Alkali_Cations	0	0	0	1	0	0	0
	Alkali_Cations	Transition_Metal_Cations	Lattice_Parameter	Composition	Unit_Cell_Angle	Volume	In_Plane_Polarization

But what about domain knowledge?

```
# Instantiate the LLM
llm = ChatOpenAI(temperature=0, model='gpt-4')

# Load tools for LangChain
tools = load_tools(["arxiv"], llm=llm)

# Initialize the agent
agent = initialize_agent(tools, llm,
agent=AgentType.CHAT_ZERO_SHOT_REACT_DESCRIPTION,
handle_parsing_errors=True, verbose=False)

# Define a function to query the LLM for causal relationships
def get_llm_info(llm, agent, var_1, var_2):
    out = agent(f"Does {var_1} cause {var_2} or the other way around?
We assume the following definition of causation: if we change A, B
will also change. The relationship does not have to be linear or
monotonic. We are interested in all types of causal relationships,
including partial and indirect relationships, given that our
definition holds.")

    pred = llm.predict(f'We assume the following definition of
causation: if we change A, B will also change. Based on the
following information: {out["output"]}, print (0,1) if {var_1}
causes {var_2}, print (1, 0) if {var_2} causes {var_1}, print
(0,0) if there is no causal relationship between {var_1} and
{var_2}. Finally, print (-1, -1) if you don\'t know. Importantly,
don\'t try to make up an answer if you don\'t know.')
    return pred

priori_knowledge = PrioriKnowledge(n_nodes=len(all_vars))
# Generate the LLM-informed DAG matrix
priori_dag = np.clip(priori_knowledge.matrix, 0, 1)
```

LLM-Informed DAG Matrix

	Alkali_Cations	Transition_Metal_Cations	Lattice_Parameter	Composition	Unit_Cell_Angle	Volume	In_Plane_Polarization
In_Plane_Polarization	0	0	0	0	0	0	0
Volume	0	0	0	0	0	0	0
Unit_Cell_Angle	0	0	0	0	0	1	0
Composition	0	0	0	0	1	1	0
Lattice_Parameter	0	0	0	0	0	1	0
Transition_Metal_Cations	0	0	0	1	0	0	0
Alkali_Cations	0	0	1	0	0	0	0

- Based on the A. Molak <https://towardsdatascience.com/jane-the-discoverer-enhancing-causal-discovery-with-large-language-models-causal-python-564a63425c93>

Put them all together!

```
# Re-run PC with Priori Knowledge
pc_priori = PC(priori_knowledge=priori_knowledge,
variant='stable')
pc_priori.learn(scaled_data)

# Replace the indices with variable names in the
DAG matrix
labels_llm_informed = [inverse_var_map[i] for i in
range(pc_priori.causal_matrix.shape[0])]
pc_dag_named_llm_informed =
pd.DataFrame(pc_priori.causal_matrix,
index=labels_llm_informed,
columns=labels_llm_informed)
```

PC-Discovered DAG Matrix with LLM assist

In_Plane_Polarization	1	0	1	1	0	1	0
Volume	0	0	0	0	0	0	0
Unit_Cell_Angle	1	0	1	0	0	1	0
Composition	0	0	0	0	1	1	0
Lattice_Parameter	0	0	0	1	0	1	0
Transition_Metal_Cations	1	0	0	1	0	1	1
Alkali_Cations	0	0	1	0	0	0	0

Key message:

- We need frameworks that allow to constrain LLM outputs to the rigid format compatible with downstream applications. This can be DAGs, knowledge graphs, JSONs, etc.
- We still need to check – by human or another ML agent
- There is (past) knowledge in numbers

What's next?

- We can perform causal analysis by mining data from sources like arXiv to establish links between film growth parameters and material properties. actionable insights
- It didn't work too well – but its possible!
- Refining queries (human heuristics), adding physics, and so on!

